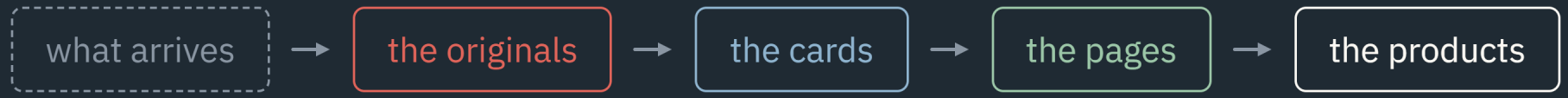
DTU · SEMINAR ON AI IN RESEARCH · 3 NOVEMBER 2026

Working with an AI agent in research

How I work, and how you can start.

Kevin Smith · DTU Construct (ES)



WHERE I STARTED

A few months ago I could not program

Until August 2026

- I did not know Python
- I had barely used VS Code
- I did not know how to clone a repository from GitHub

14 August 2026

my first repository

I describe what I want, mostly by voice, and read what comes back. The agent writes the code.

HOW I GOT HERE · 1 OF 3

For two years, every file went into a Claude Project

2024 · Claude Projects → 2025 → April 2026 → August 2026

- Claude wrote better than the other tools
- A Project: files that every chat in it can read
- I gave it everything: every lecture and meeting, transcribed
- Then the project grew, and the answers got worse

If you use Claude Projects today

this is where I was a year ago

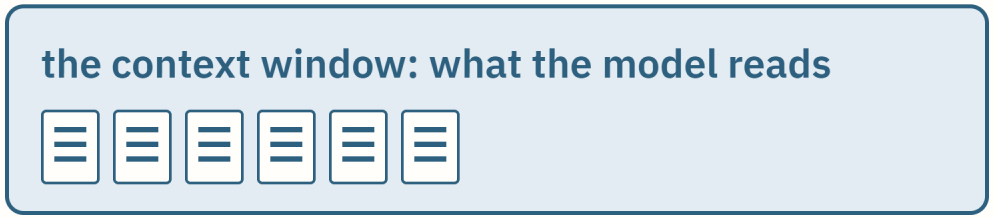
Our InnoExplorer's meetings alone

56 transcripts since March 2025, about 770 000 words

Why the answers got worse

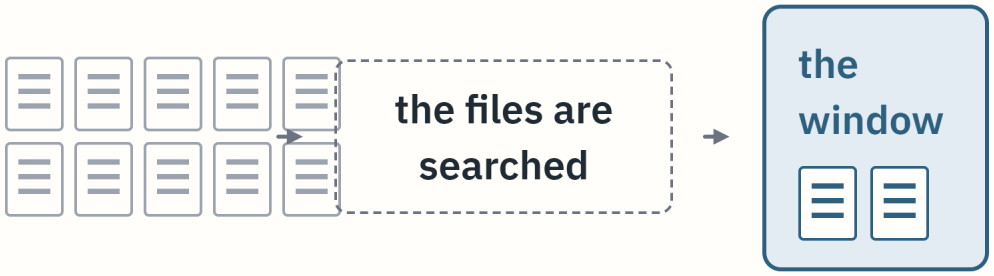
2024 → **2025 · the answers get worse** → April 2026 → August 2026

A small project



The model reads every file.

A large project

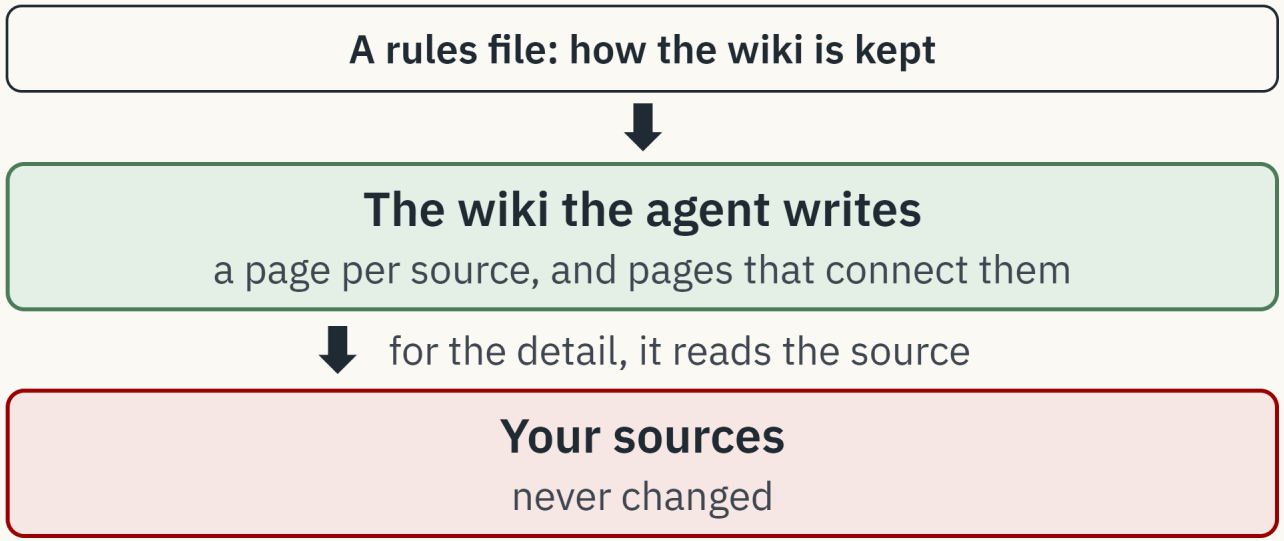


Only what the search finds reaches the model. The more it holds, the less carefully it reads.

The lesson: choose what the model reads.

Karpathy's LLM wiki: the agent writes the wiki

2024 → 2025 → **April 2026 · Karpathy's wiki** → August 2026



What my material made me add

- People talking, not papers: a card per meeting
- Who said what, and when
- What replaced what, never deleted
- My own reasoning, dictated

First I over-engineered it. Then I kept only what I needed.

THE PREMISE

Most of what we know is not in our documents

Documents

papers, applications, reports, slides, data

Already files

Conversations

lectures, meetings, calls, e-mails

Files only if recorded

Our minds

the reasoning, the why

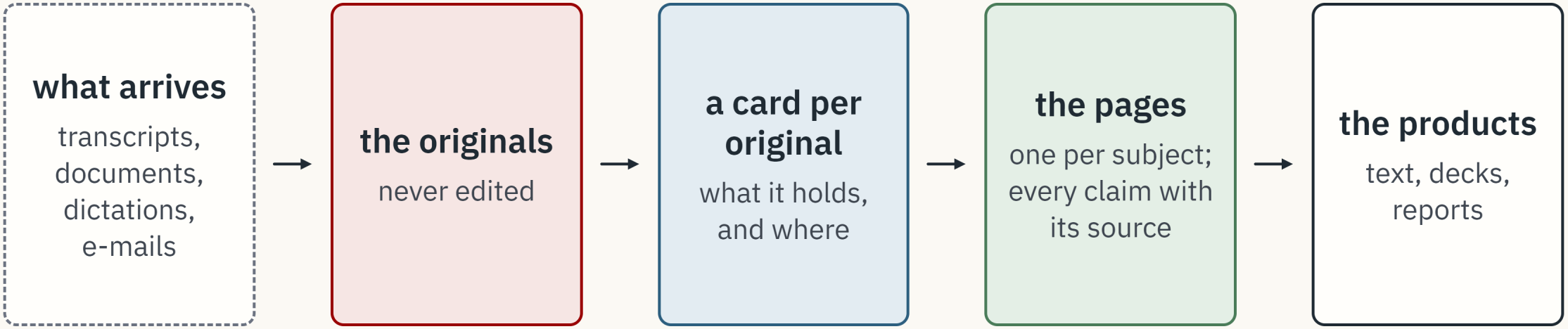
A file only if you say it aloud

Give the agent the whole of what you know.

THE METHOD IN ONE PICTURE

Everything flows one way

You: transcribe · dictate · deposit · judge the product



The agent: reads · writes · builds · records

The agent’s memory is the folder, not the conversation.

1

Three habits before any tool

Transcribe. Dictate. Deposit.

1 · Three habits

2 · The knowledge base

3 · Data security and the rules

4 · How to start

HABIT 1

Transcribe every lecture and meeting

A lecture or a meeting on Teams

Download the Teams transcript (.vtt format)

A meeting in a room, or one you don’t organise

Meetily Pro on the laptop; nothing uploaded

A recording on a conference microphone or a phone

Two free NVIDIA models, on your own computer

An e-mail that bears on the project

Save it beside the rest, as an .eml file

Everything goes into the inbox of its knowledge base. Tell everyone you record.

HABIT 2

Dictate: ideas, reactions, corrections, at length

hold a key → **talk** → **release** the text lands where the cursor is

- Wispr Flow, with a personal dictionary: your names and terms, in Danish and English
- Into a file in the inbox, or straight into the agent’s chat
- A later dictation beats an earlier one
- Switch off “Improve the model for everyone”

150 000

words dictated for one course in seven weeks

HABIT 3

Deposit, never edit

The inbox

what arrives

→

The originals

kept as they arrived: never edited, renamed or deleted; a new version beside the old

A small program refuses any change to an original

an agent deep in a long task forgets what the instructions said

Two habits for the inbox

- E-mails go in as .eml files
- E-mails of one thread get the date in their names

"Where does this come from?" is still answerable a year later.

YOUR TURN · ROUND 1 OF 3 · ABOUT 5 MINUTES

1 How much of your reasoning is in no document?

Is that what an AI tool missed when it wrote for you?

2 What would stop you recording or dictating that reasoning?

Consent, the cloud, the habit?

One question at a time, for the whole room. Objections are welcome.

2

The knowledge base, and what it produced

One rule, a round of work, a spin-out’s memory, a correction made a rule.

1 · Three habits

2 · The knowledge base

3 · Data security and the rules

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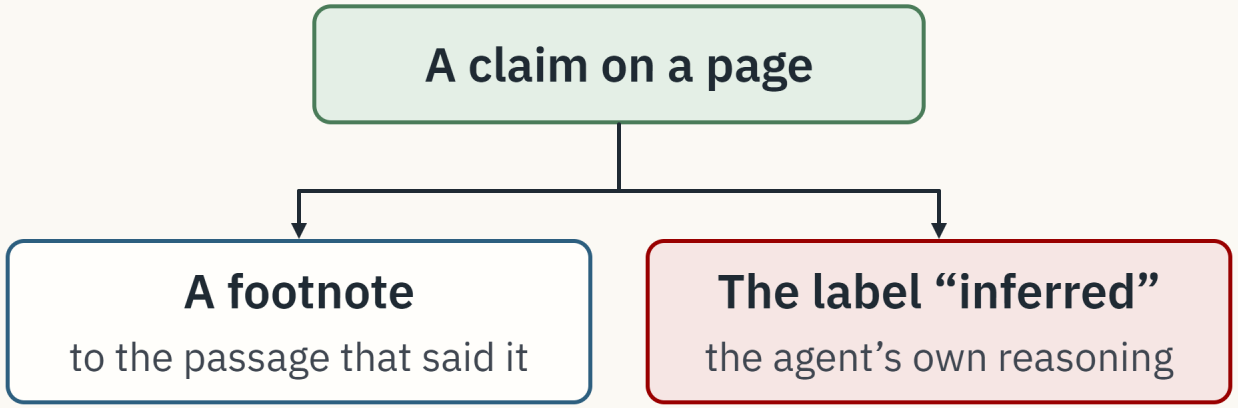
THE RULE THAT MATTERS MOST

Every claim says what it rests on



- Stated and inferred are never merged
- Superseded, never deleted
- A number read from a drawing stays inferred until a person checks it

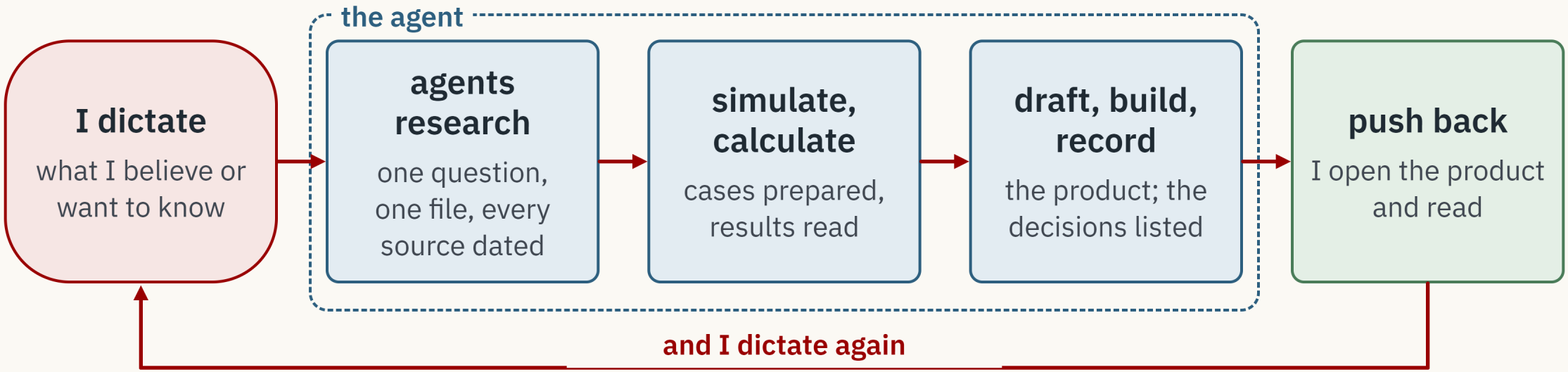
Every number carries its basis

measured · observed · design · simulated · benchmark · extrapolated · estimated · stated for a funder · judgement

There is no third option.

THE LOOP

One round: dictate, research, build, push back



"until the results get to a place where I feel comfortable with the idea and I get some pushback, and then I'll dictate again"

WHAT IT PRODUCED · A SPIN-OUT'S MEMORY, SINCE MARCH 2025

One memory for a spin-out

Every meeting and interview
transcribed since March 2025

E-mails, documents, dictations
deposited beside them

Research notes by agents
the science, the tools, commercial aspects, funding, contacts

Archive drawings
turned into the inputs of a digital twin

One memory

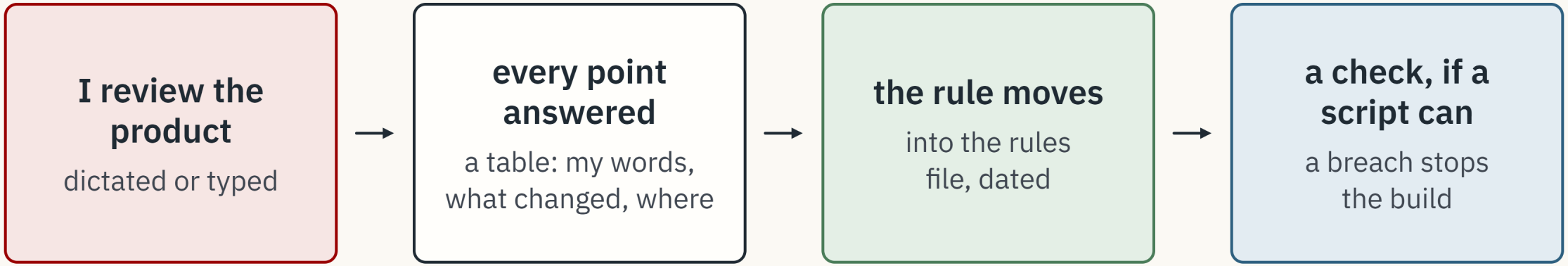
every claim says who said it, and when

Before a meeting

"What did we decide about this, and why? What is still an assumption?"

HOW A CORRECTION BECOMES A RULE

Not in the chat’s memory. In files, the same day.



Read by every later session, on every machine.

WHAT THE AGENT WRITES FOR ME

Making the agent write for a newcomer

What it wrote

“Server mode exits after ~10 min offline; phone loses session.”

What I asked for

“If my home computer is offline for about ten minutes, the program that links it to my phone stops, and the phone loses the conversation.”

1

Tell it who you are

your field, and the fields it must explain

2

Write it down once

in a file it reads at every start

3

Ask again

one command: explain that for a newcomer

IDA ICE · 27 OCTOBER 2026

A whole year simulated by one command, unattended

One command
from the agent’s shell

→

IDA ICE, batch mode
on a copy of my model

→

A full year of results
in about five minutes

One run at a time, at my word
never a loop of the agent’s own

⚠

Is a run at my request “automation” under the licence?
a question to EQUA, drafted

WHAT IT PRODUCED · ACROSS MY WORK

What I use it for

Literature
Online sources read and dated; a hypothesis tested

Data
Public data fetched and analysed; quick estimates

Simulations
Inputs and boundary conditions found; runs investigated

Administration
Funding queries, reports, e-mails, posts

Teaching
Decks, notes, assignments, exams; a textbook renewed

Course handover
A successor starts from the course's memory

Books
Open review; a version for each audience

Student feedback
Not yet: written out from my dictated points

Each product states that an agent helped make it.

YOUR TURN · ROUND 2 OF 3 · ABOUT 6 MINUTES

1 **Which task would you hand to an agent first, and which never?**

2 **What would you need to see before you trusted what it wrote?**

One question at a time, for the whole room. Tell me where this would fail.

3

Data security, privacy and the rules

What the agent can reach, what the guidelines say, what I do, and where they differ.

1 · Three habits
2 · The knowledge base
3 · Data security and the rules
4 · How to start

WHAT AN AGENT ON YOUR MACHINE CAN REACH · IT WORKS AS YOU

What it reads leaves the machine

Your privileges

It reads and writes files as you; in auto mode, without asking

The network

What it reads is sent to the provider, with your prompts and its answers

The terms

A personal plan: consumer terms. Team, Enterprise or the API: a data agreement

Rules are context

Only a hook, a deny rule or a sandbox stops an action

SWITCH TRAINING OFF, PRODUCT BY PRODUCT

Where the setting is

Product	The setting
Claude, and Claude Code on the same login	“Help improve Claude”: Settings, Privacy
ChatGPT	“Improve the model for everyone”: Data controls
Gemini	“Keep Activity”
Copilot (consumer)	“Training on conversation activity”
Wispr Flow	“Improve the model for everyone”: Data and Privacy
Meetily	none needed: it runs on your computer

“Off” stops future training. It does not undo the past. Read the toggle yourself.

WHERE MY APPROACH MAY NOT ALIGN WITH BEST PRACTICE

Three gaps, and what would close each

The terms of the account

A personal plan, and DTU wants a data agreement

↓

Ask DTU which plan

and which terms a researcher's agent should run under

Personal data the agent can read

Names in the transcripts of meetings

↓

Strip the names at ingest

and keep the agent out of the originals

The agent's reach

It reads whatever you can read

↓

A rule that asks first

before the agent reads DTU's OneDrive or drives

4

How to start

What to learn from, what I can hand you, and what I tell people to do.

1 · Three habits

2 · The knowledge base

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LEARNING MATERIAL

A few reliable sources are enough

IBM Technology

YouTube · @IBMTechnology

Ten-minute explainers at a whiteboard: how models, agents and retrieval work

AI Explained

YouTube · @aiexplained-official

Reads the papers behind every major release; keeps his own private test of reasoning

Matt Pocock

YouTube · GitHub · aihero.dev

Plan first, in writing; then small steps that you review

In 2025 I was told AI had peaked. These sources showed what was coming.

IBM TECHNOLOGY · YOUTUBE.COM/@IBMTECHNOLOGY

Five videos to start with

Start with these five	Presenter
AI Simplified: 6 Concepts You Need to Know	Jeff Crume
AI Model vs Agentic Harness	Martin Keen
Skills vs MCP vs RAG vs Memory	Martin Keen
Context Engineering for AI Agents	Martin Keen
Understanding AI Agent Hallucination	Brianne Zavala

Presenters to look for

Martin Keen and Jeff Crume

TAKE IT HOME

Everything I recommend, in one folder

Getting started

the four tools, step by step, for someone new to a terminal

A knowledge base, ready to fill

the folders, a rules file with blanks, the guard for the originals

The writing kit

tell the agent once who you are and what to explain

Prompts and sources

my prompts, and the videos to start with

Open the folder in VS Code, open Claude Code, and say: “Read START-HERE.md and set me up.”

WHAT I TELL PEOPLE TO DO

In this order

1. Transcribe every lecture and meeting. Tell people.
2. Dictate, with a dictionary. Never edit the transcript.
3. One inbox per project. Deposit, never edit.
4. Write the rules into the file the agent reads first.
5. Settle the terms of the account; DTU’s folders behind a prompt.
6. Budget for the top plan, as an instrument.
7. Judge the product, not the plan.
8. Ask the knowledge base before every meeting.

Discussion

Your questions and objections: consent, the terms of the account, who checks what the agent wrote, where this would fail.

For the section: shall we map together how our courses fit and overlap?

1 · Three habits

2 · The knowledge base

3 · Data security and the rules

4 · How to start

B

Backup slides

The detail behind the talk, for questions.

THE LAYERS

Who may write what

Layer	Holds	Written by; changes?
raw/ · the originals	every source under its identifier	a person, through the deposit tool; never
sources/ · the cards	one card per source: an index, not a replacement	a script, or an extraction agent; regenerable
kb/ · the pages	pages by type: primer, concept, decision, evidence, method; the registers	the integration agent; yes, footnoted
products	text, decks, assignments, the book	scripts; rebuilt after every correction

WHERE IT COMES FROM

Karpathy's pattern, and the additions

The pattern, April 2026

- immutable raw sources
- a compiled Markdown layer the agent reads and writes
- a schema file that governs how it may write
- three operations: ingest, query, lint

The additions

- cards: a transcript is mostly not signal
- provenance and grading: people arguing disagree with each other
- primers: the exposition an expert gives once at a whiteboard
- the rationale register: my reasoning, by argument
- precedence: later wins; colleague material is context
- hooks, not prose; a resume point and a log

THE RATIONALE REGISTER

“I’m dictating so much where I explain reasoning and rationale. This register captures all that … a lot of this rationale gets lost most of the time in our day-to-day activities. It’s part of the reason why this should achieve better performance: that rationale is there, and your outputs are not going to be counter to it.”

The author, 29 October 2026. The course’s register: about 76 000 words in seventeen arguments, each entry with the dictation and paragraph it came from.

SHORT SESSIONS; THE BASE IS THE MEMORY

One screen in, one block out, the next session clean

- At the start: the resume point, the rules file, the last log block
- At the close: a log block, the resume point rewritten, a commit
- Never read whole: the registers are searched; long reads go to a sub-agent

The Claude Code team, April 2026

“When you start a new task, you should also start a new session.”

OpenAI, 2026

“The most important technique was durable project memory.”

B

The course, in seven weeks

Six lectures, a class text, three assignments, six decks. And what I had to correct.

THE COURSE · 29 AUGUST TO 12 OCTOBER 2026

What one lecturer and one agent produced

222
originals deposited

75 000
words of class text, in three parts

6 × 19
lecture decks and their versions

3
assignments with solution sets

157
reading notes, one per original

300
commits; four autonomous nights

CORRECTION 1 · 20 SEPTEMBER

“Not how I sound”: the register of the text

What came back

- draft banners and retrieval dates in a student text
- physics in sentences instead of an equation and an example
- cut to a word limit, not checked for cohesion
- a voice that imitated my dictation

The rule that came out

- literal and plain, for second-language readers
- the referent named; calculations displayed
- no length limit; no idiom; no imitation
- provenance in a companion folder, kept for the book

CORRECTION 2 · 23 TO 27 SEPTEMBER

A slide is a cue, not a document

The fault, in my words	The rule
“far too much text, spoken rather than written”	a title of 10 words, a bullet of 14; the reasoning in the notes
“cryptic … details instead of big picture”	one idea per slide; the text pointed to, not compressed
“colloquialisms … replicating my tone”	plain clauses; the terms in force; units on every quantity
60 slides, the last block skipped	about 40; my measured pace sets the size
“the plots and graphics are all distorted”	my compositions rebuilt by code at their size

B

Simulation and agents

The agent reads the whole program. What it may run is a licence question.

IDA ICE

A model is text. The agent reads all of it.

- Inputs: zones, the AHU macro, setpoints, schedules, the climate
- Results: the hourly files, read by scripts
- Weather files: the header and the statistics
- Component models: the shipped source, read for what a fan computes

36

models deposited, a one-page card each

BACKGROUND AGENTS AND NIGHT SHIFTS

Research fanned out; idle nights used

Background research

- one question, one agent, one file
- the address and date of every source
- 8 to 14 at once, back in 10 to 20 minutes

A night shift

- what needs no word from me, and is reversible
- a commit after every unit
- at the close: the minimum questions, a recommendation each
- never: a simulation, an account, an install, a rule

The book

A living, openly reviewed, versioned work, published outside the journals.

WHY PUBLISH IT MYSELF

“Publishing has become a very democratic process. Anyone can do it, and there’s no reason why publishers should take huge amounts of money or own your copyright when we have all the tools to publish ourselves, including the review, versioning, and integrating things that we wouldn’t be able to integrate into a lot of normal publishers’ platforms, like visuals, interactive plots, and such.”

The author, 29 October 2026

THE ROUTE

Pieces, with nothing to operate

Public or nonprofit: the spine

Zenodo · a DOI per version | Software Heritage

PREreview | ORCID · DataCite

DTU Orbit · DTU Data

Open source: the engine

Quarto | Pandoc | Markdown masters

Commercial, each with an exit

GitHub Pages | Hypothesis | Claude Code

- CC BY 4.0: attribution required; AI ingestion permitted
- The review record: every version frozen with a DOI
- Open contribution through the gap register
- Stable anchors: a claim keeps its identifier
- The knowledge base as data: a server a chat agent can query

WHERE THE BOOK STANDS · 29 OCTOBER 2026

A prototype, not a release

4
candidate chapters of Part I

19
equation records, each example reproduced by a test

114
claim records; 148 evidence records with a stance

33
gaps registered: the invitation to contributors

2
profiles from one source: student and researcher

78
numbered assumptions awaiting one dictation

WHAT THE GUIDELINES SAY, AND WHAT I DO

The guidelines, and what I do

The guideline	What I do
DTU: no personal or sensitive data in prompts; Claude “not assessed”	Projects outside OneDrive; DTU’s folders behind a prompt; no credentials in a repository
GDPR: a transcript with names is personal data; tell the people recorded	Teams tells everyone; I say so in a room; nobody named in the products
The Commission (2026), the AI Act, the Danish Code: protect unpublished work; declare AI use; a person stays responsible	Private repositories; a hook protects the originals; I review and sign every product; the book will carry a statement

WHERE THE CAUTION CAME FROM

The model's default, released by hand

What the first drafts said

"People appear as P-nn." "Residents are never named." In the agent's first schema, before any word of mine.

How it travelled

Each new base was built in the pattern of the last, and the caution went with it.

What I released, by hand

Direct names; per-flat data in scope; public register data; public names. None propagated on its own.

Set the policy on day one: what the base holds is yours to decide.

WHY IT PERFORMS

Five mechanisms

1
The folder is the memory
not the conversation

2
Provenance
a footnote or “inferred”; no fabrication

3
The reasoning is in the corpus
from the dictations

4
Corrections compound
a rule in a file, the same day

5
The product is judged
rendered, and read

Claimed: throughput, my verdicts, faults that stopped recurring, traceability. Not claimed: a controlled comparison.

CUTTING EDGE OR BLEEDING EDGE?

Ordinary, leading, and ahead

Ordinary now	The leading edge	Ahead, handle with care
plain files an agent reads; a rules file at the root; the wiki pattern; sub-agents, hooks, skills	provenance as a hard rule; grading; supersession; extraction separate from integration; the rationale register; the night shift	claim and evidence records as the data layer of a published book, queryable by agents

Longer contexts do not remove the layers; memory features are per tool; plain files move between models.

AGAINST THE LEADERS, SEPTEMBER AND OCTOBER 2026

Aligned, different, and not addressed

Aligned	Differs	Not addressed by anyone
the base as the memory; the hook; skills; sub-agents for parallel work; the product rendered	the rules file too long; “end before compaction” stricter than asked; nights without a finish line	immutable originals; the inferred label; the registers; dictation; the deck as an artifact

Overkill by their measure: the length of the rules file. The rest is what they recommend, or what research needs.

PROJECT ADMINISTRATION · SPRING 2026, IN A CLAUDE PROJECT

A funding query, answered from the rules

The query
a partner's budgeted hourly rate, questioned

→

What was read
the funding rules, the collaboration agreement, the partner's annual report

→

What was proposed
actual salary rates, with overhead as the rules allow

→

What went out
the e-mails to the partner and the funding body

Then: a Claude Project
documents uploaded by hand, into one chat

Now: a knowledge base
rules, agreement and decisions already there, each with its source

THE BUDGET

A research tool, not a chat app

- About 190 euro a month for the top plan
- Written into project budgets, as an instrument
- VS Code, git, Python, pandoc: free
- A small always-on machine at home for the nights

Prices

Last year’s top model costs a quarter per token today; the newest does not. Plain files move between models.

IS IT RIGHT FOR YOU?

Three tests

- Your corpus is bigger than your memory of it
- Most of what you know comes out when you talk
- You will read the product and say what is wrong, every time

Three habits, one discipline, an account whose terms fit the data, a subscription. No developer.